

Table 1 Performance of various models on GST-Bench across three core competencies: **Object Localization**, **Self Localization**, and **Scene Structure Understanding**. The **Avg.** is the arithmetic mean of all 12 subtask scores. The highest, second-highest, and third-highest scores in each column are marked in light red, light orange, and light yellow, respectively, excluding Qwen3-VL-8B (fine-tuned), which is fine-tuned on our proposed training set GST-Train. ED = Ego. Direction, EDist = Ego. Distance, GP = Global Position, Pos = Top-Down Position, Ori = Top-Down Orientation, TDS = Top-Down Selection, Traj = Trajectory Selection. v / s = visual / semantic modality

Method	Object Localization					Self Localization		Scene Structure Understanding					Avg.
	ED _v mAcc	EDist _v MRA	GP _v mAcc	ED _s mAcc	EDist _s MRA	GP _s mAcc	Pos mAcc	Ori mAcc	TDS _E Acc	TDS _M Acc	TDS _H Acc	Traj Acc	
Proprietary Models													
Gemini-2.5-Pro	18.60	42.00	31.80	23.08	35.40	32.40	30.47	21.52	97.69	80.00	41.01	37.44	40.95
Gemini-3-Pro	21.73	25.35	42.23	22.11	27.39	43.67	37.47	16.89	98.15	82.73	52.25	42.18	42.68
GPT-5	27.98	27.35	34.28	31.09	27.39	43.23	22.33	16.74	97.69	79.55	43.26	39.34	40.85
GPT-4o	19.64	10.09	22.36	13.46	10.13	25.24	24.03	15.99	81.02	71.36	45.51	25.12	30.33
Seed1.8	16.37	12.43	27.96	20.35	13.32	36.07	28.45	17.79	99.07	67.27	37.64	31.75	34.04
Open-Source Models													
LLaVA-OV-1.5-4B	17.41	4.17	14.28	17.63	4.34	18.86	20.23	16.29	49.54	37.27	27.53	25.59	21.10
LLaVA-OV-1.5-8B	17.71	5.52	14.22	18.11	3.94	10.83	13.33	16.89	52.78	33.18	21.91	26.54	19.58
Qwen3-VL-2B	17.41	4.13	9.73	16.19	6.55	18.08	20.93	18.39	88.89	25.91	21.35	24.17	22.64
Qwen3-VL-4B	15.33	4.09	17.46	16.66	6.55	12.66	19.61	18.09	94.44	34.09	22.47	26.07	23.96
Qwen3-VL-8B	18.75	5.57	17.35	12.50	11.59	13.01	24.81	17.64	96.76	42.27	25.28	25.12	25.89
Qwen3-VL-32B	16.67	13.96	18.05	15.22	16.19	26.55	24.03	17.64	94.44	60.45	32.58	29.38	30.43
InternVL3.5-2B	18.45	2.70	9.50	18.11	4.56	13.97	12.25	17.34	59.72	29.09	25.84	25.12	19.72
InternVL3.5-4B	15.63	4.96	8.91	12.02	5.40	5.15	20.62	16.29	66.67	30.91	25.28	26.07	19.83
InternVL3.5-8B	17.71	5.57	19.06	16.67	8.36	14.85	18.84	13.90	75.00	35.00	25.84	25.12	22.99
InternVL3.5-38B	21.28	11.13	20.29	20.03	10.44	33.10	22.09	20.63	75.00	68.64	36.52	29.38	30.71
NVILA-8B	18.15	7.22	15.87	16.02	7.52	26.64	6.67	16.44	79.17	34.55	29.78	25.59	23.64
NVILA-15B	15.47	4.74	8.85	16.66	3.10	10.13	8.06	17.79	74.07	40.00	32.02	22.27	21.10
Embodied-Understanding Models													
Cosmos-Reason2-2B	15.18	3.91	8.38	14.10	7.65	8.47	17.75	16.74	78.70	24.09	26.40	18.48	19.99
Cosmos-Reason2-8B	13.39	6.96	10.86	16.83	7.61	10.57	19.77	14.50	84.26	28.64	23.60	22.75	21.64
RoboBrain2.5-8B	20.24	10.39	14.04	19.23	11.28	14.06	20.31	16.74	80.56	37.27	26.97	24.17	24.61
Robix-7B	17.41	12.35	24.13	18.43	12.52	21.83	22.95	15.99	88.43	53.18	35.96	25.59	29.06
Robix-32B	18.90	24.87	21.95	18.91	20.18	21.05	22.48	17.94	84.72	43.64	30.90	25.59	29.26
Baseline													
Random Guessing	14.88	23.22	15.04	17.31	20.40	14.15	16.51	15.55	23.15	27.73	24.72	27.49	20.01
Human Level (20 samples/task)	75.00	41.50	93.00	70.00	41.50	89.00	84.00	85.00	100.00	95.00	85.00	90.00	79.08
Ours (Fine-tuned on GST-Train)													
Qwen3-VL-8B (fine-tuned)	53.72	39.70	48.97	49.36	31.95	57.64	40.55	44.39	99.07	88.18	37.08	51.66	53.52